



# TaskPilot

AI Project Management Chat Module

←  **Technical Report**

Functional Documentation & System Design

Platform

Stack

AI

Event

Version

COMPETITION SUBMISSION READY

Chat

Tasks

Dashboard

Team

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# Abstract

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## Executive Summary

**TaskPilot** is a production-quality **AI Project Management Chat Module**: conversational control, structured task CRUD, explainable priority scoring (0–100), live progress analytics, and team slots (max. 8). It runs entirely on **localhost** with Flask + SQLite and a polished dark-themed GUI. Built-in AI guarantees reliable commands; optional GPT enriches natural-language replies.

**5**

Core  
mod-  
ules

**12+**

REST  
end-  
points

**0–**

**100**

Priority  
score

**8**

Team  
slots

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# 1

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## Introduction

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*“The best interface is a conversation that gets work done.”*

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—TaskPilot design principle

### Vision

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Modern teams need more than static todo lists. **TaskPilot** unifies *natural language*, *structured tasks*, and *visual analytics* so users plan smarter and ship faster.

**Problem:** Manual prioritization wastes time and hides deadline risk.

**Solution:** AI-assisted creation, automatic scoring, and real-time dashboards.

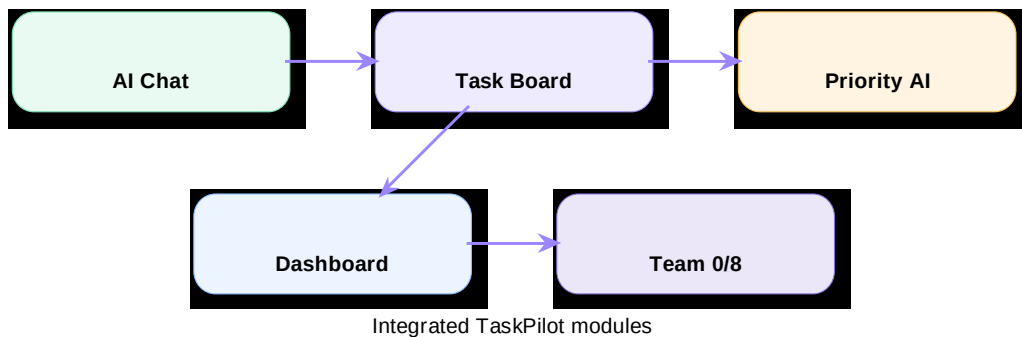
**Delivery:** Full web product at `http://127.0.0.1:5000/login/` with session auth and REST API.

### Core Objectives

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1. **Chat-first workflow**—create, list, and prioritize via conversation
2. **Transparent AI**—every score is explainable to judges and users
3. **Production GUI**—not a CLI-only prototype; complete visual product
4. **Submission-ready**—README, `.env.example`, docs, share ZIP, this report

## Functional Requirements



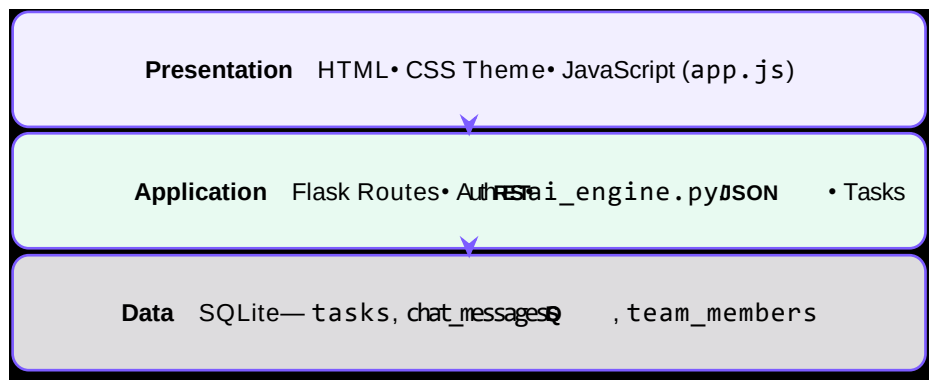
**Figure 2.1:** Functional module map

## Requirements Traceability

**Table 2.1:** Competition requirements mapped to TaskPilot features

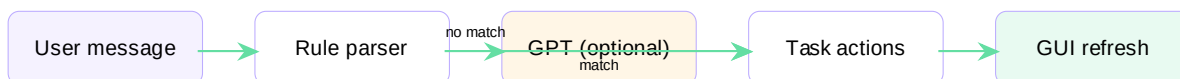
| Requirement        | Implementation                                                                 |
|--------------------|--------------------------------------------------------------------------------|
| Chat interface     | AI panel, history, quick chips, commands (add task, dashboard, top priorities) |
| Task management    | CRUD, search, filters, modal editor, deadline quick-picks, priority pills      |
| Priority scoring   | Multi-factor 0–100 score; Critical / High / Medium / Low + explain API         |
| Progress dashboard | Completion %, donut chart, status bars, priority grid, top-8 queue             |
| Team collaboration | Up to 8 members, roles, live N / 8 Occupied counter                            |
| Localhost GUI      | Flask + session cookies; dark theme; Demo Data one-click samples               |

## System Architecture



**Figure 3.1:** Three-tier system architecture

## AI Message Pipeline



## User Interface & Features

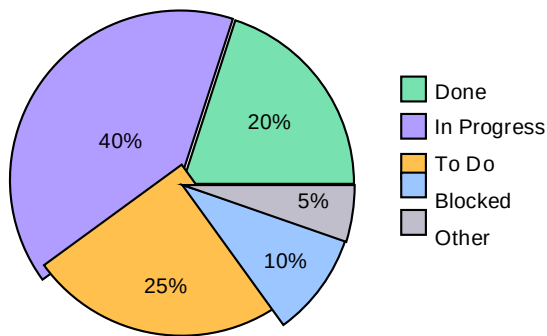


Figure 4.1: Sample task status mix (demo data)

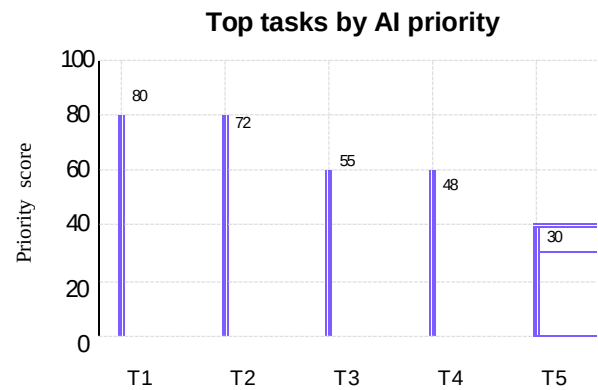


Figure 4.2: Priority score distribution

[

Login & Auth] Terminal-styled sign-in, demo mode (any email/password), session cookies, logout.

[

Welcome Hero] Personalized *Welcome back, Name!* banner with purple gradient highlight.

### Workspace Highlights

**AI Assistant**—GPT / Built-in badge, connection status, typing indicator

**Task Board**—priority-sorted cards, edit/delete, *Explain priority*

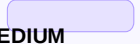
**Toolbar**—**Demo Data** (replace samples), Recalculate, AI Suggest Deadlines

### Dashboard & Team

Dashboard shows completion %, stat cards, donut chart, status breakdown, priority grid, and top-8 active queue. Team tab supports roles (Member, Lead, Developer, Designer) with **8** maximum slots.

## AI Priority Algorithm

**Table 5.1:** Priority bands (aligned with UI pills)

| Score     | Label    | Visual                                                                              |
|-----------|----------|-------------------------------------------------------------------------------------|
| $\geq 75$ | Critical |    |
| 55-74     | High     |   |
| 54        | Medium   |  |
| $< 35$    | Low      |  |

### Scoring Model

$$\text{Score} = \text{clamp}(40 + \square_{kw} + \square_{dl} + \square_{st} + \square_{ef}, 0, 100)$$

Keywords (urgent, critical), deadline proximity, status, and effort hours drive the result.



## API Reference

| Method   | Endpoint               | Description                     |
|----------|------------------------|---------------------------------|
| GET      | /login/                | Login page                      |
| GET      | /                      | Main workspace (auth required)  |
| POST     | /api/chat              | AI assistant message + actions  |
| GET      | /api/dashboard         | Stats and chart payloads        |
| GET/POST | /api/tasks             | List or create tasks            |
| PATCH    | /api/tasks/<id>        | Update task fields              |
| DELETE   | /api/tasks/<id>        | Remove task                     |
| POST     | /api/seed-demo         | Load demo tasks (replace: true) |
| GET/POST | /api/team              | Team members                    |
| POST     | /api/tasks/recalculate | Re-score all tasks              |

**Table 6.1:** Primary REST API surface

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## Installation & Configuration

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### Quick Start—Windows

```
1 python -m venv venv
2 .\venv\Scripts\activate
3 pip install -r requirements.txt
4 copy .env.example .env
5 python run.py
6 # Browser: http://127.0.0.1:5000/login/
```

#### Note

OpenAI is **optional**. All task commands work with the built-in rule engine; set `OPENAI_API_KEY` only for richer chat replies.

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## Testing & Judge Demo

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### Automated Verification

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`scripts/test_api.py` exercises health, login, seed-demo, chat task creation, priorities, and dashboard.

### Manual Checklist

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Login + welcome banner

**Demo Data** loads 5 sample tasks

Dashboard charts populate

Chat: add task ... urgent due tomorrow

Deadline quick-picks and clear

Team add/remove within 8 slots

### 5-Minute Demo Script

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1. Problem statement (30 s)
2. Chat commands + **Demo Data** (90 s)
3. Task board + AI suggest deadlines (60 s)
4. Dashboard analytics tour (60 s)
5. Team tab + wrap-up (60 s)

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## Conclusion

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**TaskPilot** is a complete, judge-ready submission: working localhost GUI, Flask backend, explainable AI priorities, professional documentation, and distributable `TaskPilot-share.zip`.

**Strengths:** Reliability, visual polish, modular Python packages, full competition checklist.

**Future enhancements:** Cloud hosting, WebSockets, Kanban lanes, email notifications.

**Thank you for reviewing TaskPilot**

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